eukaryotes have several types of DNA polymerases (Example: DNA polymerase α [alpha] and DNA polymerase δ [delta]). Some have been linked to particular functions, such as the replication of mitochondrial DNA. The termination of replication on linear eukaryotic chromosomes involves the synthesis of special structures called telomeres at the ends of chromosome.

HOTS

What will happen if single stranded binding proteins are not present during replication of DNA?

Summary

Many lines of evidence show that DNA bears genetic information. Frederick Griffith showed transformation of bacteria. Avery, Mac Leod, Mc Carty experiment and further experiment by Hershey Chase confirmed that DNA is the transforming principle.

There are two types of nucleic acid: RNA and DNA. Nucleic acids are polymers of nucleotides, joined together by phosphodiester linkages between the 5′hydroxyl group of one pentose and the 3′hydroxyl group of the next. The nucleotides in RNA contain ribose, and the common pyrimidine bases are uracil and cytosine. In DNA, the nucleotides contain deoxyribose sugar, and the common pyrimidine bases are thymine and cytosine. The primary purines are adenine and guanine in both RNA and DNA.

Erwin Chargaff rules states that A = T and G = C. Watson and Crick postulated that DNA consists of two antiparallel chains in a right-handed double-helical arrangement. Complementary base pairs, A = T and G ≡ C, are formed by hydrogen bonding within the helix. The basepairs are stacked perpendicular to the long axis of the double helix. DNA can exist in several structural forms. Two variations of the Watson-Crick form, or B-DNA, are A-DNA and Z-DNA.

Replication of DNA occurs with very high fidelity and at a designated time in the cell cycle. Replication is semiconservative, each strand acting as template for a new daughter strand. It is carried out in three phases: initiation, elongation, and termination.

The reaction starts at the origin and usually proceeds bidirectionally. DNA is synthesized in the 5′→3′ direction by DNA polymerases. At the replication fork, the leading strand is synthesized continuously in the same direction as replication fork movement; the lagging strand is synthesised discontinuously as Okazaki fragments, which are subsequently ligated by DNA ligases.

Most cells have several DNA polymerases. In *E. coli,* DNA polymerase III is the primary replication enzyme. Replication of the *E. coli* chromosome involves many enzymes and protein factors. Replication is similar in eukaryotic cells, but eukaryotic chromosomes have many replication origins.

TN_GOVT_XI_Micro_Biology_CH14.indd 268 16-02-2019 10.37.53 AM